Latent Algorithms of Indigenous Weaving Networks: Precolonial Computation and Textile Knowledge Systems

Introduction: Weaving Worlds, Weaving Code

For millennia, across the diverse cultures of the Pacific and the Andes, weaving has been far more than a craft; it is a fundamental mode of knowledge production, transmission, and expression. Textiles served as powerful agents, encoding identity, status, cosmology, and ancestral belief systems long before the advent of silicon-based computation.[1] This report delves into the intricate relationship between Indigenous weaving practices and computational principles, exploring the "latent algorithms" embedded within the structure, patterns, and knowledge systems of Pacific and Andean textiles.

Drawing upon the field of ethnomathematics, which recognizes mathematics as a cultural construct reflecting diverse practices and values [2], this analysis seeks to understand the mathematical ideas inherent in weaving – symmetry, recursion, iteration, information encoding – not as mere abstract concepts, but as integral components of living cultural traditions.[4] The objective is to surface the iterative rule systems encoded in these textiles, analyzing how complex information traveled across precolonial trade routes and kinship networks.

This exploration operates through multiple lenses: a computational pattern lens identifies latent mathematical structures [6]; an oral-knowledge lens respects living epistemologies and transmission methods [7]; a decolonial systems lens critically examines the application of Western frameworks like "algorithm" [11]; a circulation lens traces knowledge across geographies [14]; and an AI/Art translation lens considers contemporary dialogues between heritage and generative aesthetics.[16]

Crucially, this work is guided by ethical requirements demanding collaborative, consent-based engagement, treating algorithms as co-produced knowledge, and avoiding extractive metaphors [User Query Ethics]. It grapples with the inherent tension identified in the reflexive prompt: "What is lost when I try to 'formalize' a cultural practice into code? Am I listening for logic or listening to relationships?" The aim is not to reduce rich cultural practices to abstract logic, but to appreciate the sophisticated computational thinking interwoven with relationality, spirituality, and embodied knowledge within these textile traditions.

Ethnomathematics and Weaving: Culturally Embedded Mathematical Practices

Ethnomathematics provides a crucial framework for understanding the mathematical dimensions of weaving, challenging the traditional view of mathematics as a neutral, universal science.[2] Instead, it posits mathematics as a cultural construct, reflecting the specific needs, values, and practices of different social groups throughout history.[2] This perspective moves beyond simply identifying geometric shapes in artifacts; it explores the mathematical ideas embedded in activities like counting, measuring, classifying, spatial reasoning, and pattern generation as they manifest in cultural practices such as weaving, navigation, construction, and ritual.[18]

Academic mathematics is often presented as a dominant regime of truth, potentially rendering other ways of thinking about number, space, and form as inadequate.[5] Ethnomathematics, conversely, often focuses on the mathematical practices of marginalized groups, integrated into everyday life and lived cultural values.[4] However, a discursive interpretation suggests that both academic mathematics and ethnomathematics can be understood as

discourses, ways of forming possibilities, rationality, and structures, thereby challenging a strict duality or hierarchy.[4] Ubiratan D'Ambrosio, a key figure in the field, proposed a broad notion where even specialized fields like engineering or insurance mathematics could be considered forms of ethnomathematics, specific to particular cultural groups.[4]

Weaving, by its very nature, involves mathematical thinking. The creation of textiles requires planning, counting threads, managing tension, understanding spatial relationships, and executing precise, often repetitive, actions to create patterns.[19] Ethnomathematical studies highlight how weavers employ concepts like symmetry, geometric transformations (translation, reflection, rotation), linearity, and pattern analysis, often without using formal mathematical language.[19] These mathematical ideas are not abstract exercises but are deeply contextualized and connected to the weavers' everyday experiences and cultural identities.[2]

Table 1: Contrasting Perspectives: Formal Mathematics vs. Ethnomathematics

Feature	Formal (Academic) Mathematics	Ethnomathematics
Nature	Often viewed as universal, neutral, abstract	Viewed as a cultural construct, contextualized
Focus	Generalization, decontextualization, formal proof	Practices, values, needs of specific cultural groups
Origin	Primarily associated with Western academic tradition	Diverse origins in everyday activities & cultures
Representation	Symbolic notation, equations	Embodied in artifacts, practices, language, oral traditions
Validation	Peer review, logical deduction	Community practice, utility, cultural coherence
Relationship to Culture	Often separated from cultural context	Integrated with cultural values and worldview
Critique	Can be seen as hegemonic, exclusive [5]	Sometimes viewed as lacking rigor (by formalists)
Sources	2	2

Understanding weaving through an ethnomathematical lens allows for an appreciation of the sophisticated intellectual work involved, recognizing it as a form of mathematical practice embedded within specific cultural contexts. It shifts the focus from simply finding "math in" weaving to understanding weaving *as* a mathematical activity, interwoven with cultural meaning, social relations, and historical continuity.

The Computational Pattern Lens: Symmetry, Recursion, and

Encoding in Thread

Analyzing Pacific and Andean textiles through a computational lens reveals intricate systems of pattern generation, information storage, and logical structuring that resonate strongly with computational concepts. This analysis focuses on identifying latent mathematical principles like symmetry, recursion, and encoding methods, recognizing them as inherent aspects of the weaving process itself.

Symmetry: Order and Cultural Preference

Symmetry is a fundamental organizing principle in many Indigenous textile traditions. Symmetry classification, using mathematical frameworks like frieze groups (for linear patterns) and crystallographic groups (for planar patterns), serves as an analytical tool for anthropologists, archaeologists, and design historians to examine traditional patterns, indicating cultural adherence, continuity, and change.[7] Cultural groups often exhibit preferences for specific symmetries, which become characteristic features of their design systems.[7]

- **Andean Examples:** Andean textiles demonstrate sophisticated use of symmetry. Double weaving techniques, particularly prominent in ancient Peru, create complex symmetries between the front and back of a textile, often involving color reversal or translation.[8] For example, a double-woven strip might show pattern 'p p p...' on the front and its color-reversed, translated reflection 'b b b...' on the back (frieze symmetry T).[8] More complex symmetries like TG (translation + glide reflection) and TVG (translation + vertical reflection + glide reflection) were also implemented, sometimes resulting in the back pattern exactly replicating the front after a translation.[8] Multi-color weaving (triple or quadruple) allowed for even more intricate front-back symmetries, such as TR (translation + 180° rotation), where the entire color substitution corresponds to a rotation.[8] Beyond front-back relations, symmetry within the pattern itself was significant. Analysis of *tin chok* skirts from Thailand (though not Pacific/Andean, illustrating the method) showed that the alignment of vertical symmetry axes across different pattern bands could distinguish traditional versus newer weaving methods, reflecting changes in production logic.[7] Andean headbands and mantles also display symmetry in repeating motifs like snakes or geometric patterns derived from cord structures.[6]
- **Pacific Examples:** Ethnomathematical studies of Pacific weaving, such as Tlingit, Haida, and Tsimshian spruce root baskets [19], Malay *kelarai* weaving [22], Maranao *Karara* weaving [21], and Indonesian *Ulos* [20], explicitly analyze geometric designs for patterns of symmetry, rotation, reflection, and translation. Studies identify specific frieze group types (e.g., types III, VI, VII) and crystallographic patterns (e.g., p4m) present in textiles like Ulos.[20] Malay weaving analysis identifies rotational symmetry (e.g., 4-fold, 8-fold) and reflectional symmetry in patterns like *Kelarai Mata Bilis* and *Kelarai Bunga Pecah Lapan*.[22] Northwest Coast basket patterns are analyzed for units of repeat and lines of symmetry.[19] Niuean *hiapo* (tapa cloth) from the 1880s featured complex geometric designs with spirals, concentric circles, squares, and triangles arranged symmetrically.[23]

The deliberate use and manipulation of symmetry in these traditions point towards a deep understanding of geometric principles, applied not just for aesthetic appeal but potentially also for encoding cultural information or organizing social activities, as hypothesized for Andean cord patterns.[6]

Recursion and Iteration: Building Complexity from Simple Rules

Weaving is inherently an iterative process, building complex structures and patterns through the repetition of basic actions (over, under, turning, knotting). This iterative logic is closely related to the computational concept of recursion, where a process is defined in terms of itself or simpler versions of itself.

- **Pattern Generation:** Many textile patterns are generated through the recursive application of simple rules. Geometric motifs in Quechua and Aymara textiles often involve repetitive elements like zigzags and diamonds.[25] Malay *kelarai* patterns are built through repetitive intertwining movements, creating continuous, patterned, symmetrical mappings.[22] Navajo weavers use iterative algorithms like "up one over one" (one weft over one warp) to create specific angles and shapes, consciously managing effects like aliasing (jagged edges) similar to concerns in computer graphics.[26] The construction of complex patterns from repeating units, as seen in Northwest Coast basketry [19] or Maranao weaving [21], demonstrates this iterative logic.
- **Diminishing Motifs:** A clear example of recursive logic is seen in Niuean *hiapo* from the late 19th century, which featured "diminishing motifs" where design elements decrease in size from the border towards the center.[23] This suggests a recursive rule applied across scale.
- **Algorithmic Thinking:** The ability to generate intricate and diverse patterns by varying simple, repeated actions implies an underlying algorithmic understanding. The weaver executes a sequence of steps (an algorithm) to achieve a desired outcome, often holding complex patterns entirely in memory.[10] The process of weaving itself – lifting specific warp threads based on a memorized or encoded pattern to allow the weft to pass – mirrors the logic of rule-based systems fundamental to computation.

Information Encoding: Knots, Colors, and Structures as Data

Beyond visual patterns, textiles, particularly in the Andes, served as sophisticated systems for encoding and storing information, predating written scripts in the region.[9]

- **The Khipu System:** The most famous example is the Andean *khipu* (or *quipu*), a complex system of knotted cords used extensively by the Inca and preceding cultures like the Wari.[28] Khipus encoded primarily numerical data using a base-10 positional system.[6] Knots tied in specific ways (overhand, long knot, figure-eight) and positions represented digits (ones, tens, hundreds, etc.).[6] The absence of a knot could represent zero.[30] This system allowed for recording quantitative information like census data, tribute obligations, calendrical information, and potentially military organization.[6] Summation relationships often exist within khipus, where one cord sums the values on subsequent cords, aiding verification.[28]
- **Beyond Numbers:** Khipus likely encoded non-numeric information as well, although this aspect remains largely undeciphered.[28] Cord color, ply direction (S-twist vs. Z-twist), spacing, knot direction, and the hierarchical structure of primary and pendant cords are all thought to convey categorical or narrative information.[6] Recent research suggests binary encoding in cord attachment methods might relate to social structures like moieties.[28] Khipus served as mnemonic devices, aiding specialized *khipukamayuqkuna* (khipu keepers/readers) in recalling histories, genealogies, and other narratives alongside the numerical data.[9]
- **Weaving Structures as Codes:** The mathematical principles embedded in fiber structures themselves were used to generate visual codes in woven textiles.[6] Patterns resembling plied cords (with varying numbers of plies/colors and twist directions) or braids appear on Paracas mantles and headbands.[6] The visual representation of a 3D cord structure on a 2D textile indicates a sophisticated understanding of geometric projection and encoding.[6] Such patterns might have functioned as formulas or maps for organizing social activities, linking spatial arrangements (number/pairing of groups) with temporal sequences (rotation/repetition).[6]

The existence of systems like the khipu and the structural encoding in woven patterns demonstrates that Andean cultures developed sophisticated, tangible, rule-based systems for information processing, storage, and retrieval –

functions directly analogous to modern computation. The very materials and processes of fiber manipulation provided the medium and the logic for these precolonial computational systems. This tangible, embodied form of computation challenges Eurocentric notions that equate computation solely with abstract symbols or electronic machines.

The Oral-Knowledge and Circulation Lens: Transmitting and Networking Weaving Wisdom

Understanding the computational aspects of weaving requires looking beyond the finished textile to the systems of knowledge transmission and circulation that sustained these practices across generations and geographies. Indigenous knowledge systems, particularly in pre-literate societies, relied heavily on oral tradition, embodied practice, and mnemonic devices, often networked through kinship and trade.

Knowledge Transmission: Orality, Embodiment, and Mnemonic Aids

In the absence of widespread written language in precolonial Andes and much of the Pacific, knowledge, including the complex rules and patterns of weaving, was transmitted primarily through non-written means.[3]

- **Oral Tradition:** Storytelling, chants, genealogies, and myths served as primary vehicles for transmitting history, cultural values, social norms, and practical knowledge.[9] Weaving knowledge was often passed down through oral instruction, demonstrations, and the use of existing fabric samples as guides.[7] Elders played a crucial role as keepers and transmitters of this wisdom.[10] Storytelling itself can be interwoven with the practice, as seen with Māori weavers sharing life experiences while learning cloak-making [34], or weaving being used as a structure for storytelling in research.[33]
- **Embodied Practice:** Weaving is a full-bodied, embodied practice requiring dexterity, strength, concentration, and memory.[10] Knowledge is acquired through doing, through the "finger choreography" and physical engagement with the loom and materials.[10] Weavers often memorize vast numbers of complex patterns.[10] This embodied knowledge is deeply connected to place, materials (wool, dyes from local plants), and community relationships.[10] The act of weaving itself becomes a mode of knowledge production and transmission.[17]
- **Mnemonic Devices:** To aid memory and ensure accuracy in complex systems, various mnemonic devices were employed.
  - *Khipus:* Beyond their function as data records, khipus served as powerful mnemonic aids for recalling narratives, histories, and genealogies associated with the encoded numbers and categories.[9] The physical structure of knots, colors, and cords prompted recall for trained specialists.[9]
  - *Pictographs & Symbols:* While distinct from weaving knots, other Native American mnemonic systems like pictographs used conventional, selective, and sequential visual cues to represent complex information or narratives.[37] The logic of these systems, like khipus, involved mental artifacts designed for memory recall.[37]
  - *Gestures & Chants:* While less documented in the provided snippets specifically for weaving transmission, oral traditions often involve gestures and chants as mnemonic components.[9] The rhythmic nature of weaving could easily lend itself to accompanying chants encoding patterns or sequences.
  - *Knots in Weaving:* Specific knot types or sequences within the weaving process itself could serve mnemonic functions, signaling pattern changes or counts.[38]

These transmission methods highlight that Indigenous knowledge systems are often holistic, integrating practical

skill, cultural meaning, social context, and embodied experience, rather than separating abstract knowledge from its application.[39]

Networked Knowledge: Diffusion Across Space and Time

Textile motifs, techniques, and the knowledge embedded within them did not exist in isolation but circulated across precolonial landscapes through trade, migration, kinship ties, and exchange networks.

- **Archaeological Evidence (Lapita):** The Lapita cultural complex (c. 1600-500 BCE) provides significant archaeological evidence for the diffusion of cultural traits, including distinct pottery styles, across Island Melanesia and into Western Polynesia.[14] Lapita pottery is characterized by intricate dentate-stamped designs, often featuring geometric motifs, circles, and sometimes stylized faces arranged in bands.[14] The widespread distribution of this pottery style across vast ocean distances points to sophisticated seafaring capabilities and established networks of interaction.[15]
- **Textile Connections:** While direct evidence for Lapita textiles is scarce, it has been theorized that the complex geometric patterns on Lapita pottery may have originated from, or been transferred to/from, less durable materials like *tapa* (bark cloth) or woven mats, or even tattoos.[14] The designs themselves, with their structured, repetitive elements, share characteristics with later Polynesian art forms, suggesting deep historical roots and transmission of aesthetic principles.[42] The decline of decorated Lapita pottery and the rise of Polynesian plainware around 2800-2000 years ago marks a significant cultural transition, but continuity in underlying design structures or techniques related to textiles is plausible.[14]
- **Andean & Pacific Regional Variation and Exchange:** In the Andes, distinct regional weaving styles, color palettes, and designs existed, often specific to individual villages or communities.[36] Four-cornered hats differentiated Wari and Tiwanaku elite [1], checkerboard patterns signified Inca military prowess [1], and specific hat styles indicated marital status among Chinchero women.[36] Similarly, Aymara textiles had unique regional color combinations.[25] Despite this local specificity, evidence suggests exchange and influence. Inca *cumbi* textiles, though centrally regulated, were produced across the empire.[1] Colonial-era textiles often blended Indigenous, European, and even Asian motifs, reflecting ongoing intercultural exchange.[1] The spread of specific techniques, like double or multiple weaving in the Andes [8], or motifs across different cultural groups points to interaction and knowledge sharing. In the Pacific, the presence of similar tapa-making techniques (pounding mulberry bark) across Polynesia, with regional variations in decoration methods (felting, staining, stamping, stenciling), indicates shared origins and subsequent diversification.[23] Tapa itself played a crucial role in ceremonial exchange and reciprocity networks.[23]

The diffusion of textile knowledge was not merely a passive spread of patterns but an active process embedded within social relationships, trade partnerships, and kinship obligations. Knowledge flowed along these established human networks, adapting and transforming as it moved, demonstrating the interconnectedness of communities across vast distances in the precolonial Pacific and Andes. This networked view challenges simplistic models of isolated cultural development and highlights the dynamism of Indigenous knowledge systems.

The Decolonial Systems and Ethical Lens: Critiquing Formalization and Centering Indigenous Protocols

Applying computational frameworks to Indigenous weaving necessitates a critical engagement with the potential pitfalls of formalization and a steadfast commitment to ethical protocols grounded in Indigenous epistemologies

and data sovereignty. Failure to do so risks perpetuating colonial dynamics of extraction, misrepresentation, and the imposition of external logics onto culturally embedded practices.

Critiques of Formalization: Reductionism and Epistemic Violence

The attempt to translate the complex, embodied, and relational knowledge of weaving into formal systems like algorithms or mathematical notation carries significant risks.

- **Reductionism and Decontextualization:** Western mathematics and computation often prioritize decontextualization and generalization.[11] Applying concepts like "algorithm" or formal logic can strip weaving practices of their rich cultural, social, spiritual, and ecological context, reducing them to abstract structures.[11] For cultures where knowledge is inherently relational, believing "everything belongs and exists in its relationship with everything else," removing practice from context can render it meaningless.[11] Focusing solely on geometric patterns might overlook the stories, prayers, or ecological knowledge woven into the fabric.[35]
- **Imposing External Frameworks:** Concepts like "symmetry," "algorithm," or even "computation" are products of specific Western intellectual traditions.[44] Imposing these terms can flatten the nuances of Indigenous conceptual systems and ways of knowing.[46] It risks framing Indigenous knowledge merely as a precursor to, or variant of, Western science, rather than recognizing it as a complete and valid system in its own right.[11] Formal logic itself has been critiqued for historically legitimizing dominant discourses and suppressing alternative forms of reasoning.[44]
- **Epistemic Violence:** Uncritically applying formal models can constitute a form of epistemic violence, overriding or devaluing Indigenous ways of knowing, being, and doing.[33] It can reinforce colonial power dynamics where Western frameworks are positioned as the universal standard for validating knowledge.[11] The very notion of automated "intelligence" in AI, for instance, often reflects a singular cultural definition of knowing derived from European colonial values, creating an epistemic gap.[45]
- **Ignoring Embodied and Relational Knowledge:** Indigenous epistemologies often emphasize knowledge gained through embodied experience, sensory engagement, intuition, and relationships with human and non-human kin (land, plants, animals, spirits).[13] Formalization struggles to capture this holistic, sensory, and relational quality of knowledge inherent in practices like weaving.[35] Knowledge is often understood not as abstract data, but as part of indivisible cultural wholes encompassing practices, values, and worldviews.[39]

Therefore, any attempt to model or formalize aspects of Indigenous weaving must proceed with caution, reflexivity, and in deep collaboration with knowledge holders, prioritizing cultural co-interpretation and ensuring that the chosen frameworks do not distort or diminish the original meaning and context.[46]

Table 3: Contrasting Approaches to Analyzing Indigenous Weaving

Feature	Formal / Structural Analysis	Cultural / Relational Analysis
Primary Focus	Mathematical patterns, symmetry, algorithms, structure	Cultural context, meaning, relationships, process
Key Concepts	Symmetry groups, recursion, encoding, geometry	Epistemology, cosmology, kinship, embodiment, place

Nature of Knowledge	Abstract, quantifiable, decontextualized	Holistic, qualitative, contextualized, relational
Methodology	Mathematical modeling, computer vision, pattern analysis	Ethnography, oral history, collaboration, sensory methods
Potential Outcome	Identification of logical rules, structural similarities	Understanding of cultural significance, worldview, practice
Risk	Reductionism, misrepresentation, decontextualization [11]	Potential lack of generalizability (from formalist view)
Goal	To "decode" underlying logic	To understand weaving as embedded practice [33]
Sources	6	10

Ethical Protocols: Consent, Sovereignty, and Respectful Engagement

Engaging ethically with Indigenous knowledge, particularly in research involving cultural heritage like weaving patterns, requires adherence to specific protocols that go beyond standard academic practices. These protocols center Indigenous rights, control, and perspectives.

- **Free, Prior, and Informed Consent (FPIC):** This is a fundamental prerequisite. Consent must be obtained from the traditional owners (often collectively, not just individuals) *before* any research, recording, study, use, or display begins.[51] Consent must be *free* (without coercion), *prior* (obtained in advance), *sustained* (an ongoing process, not a one-off event), and *informed* (based on a full understanding of the project's aims, methods, risks, and benefits, communicated transparently).[52] This principle is enshrined in international conventions and ethical guidelines.[52]
- **Indigenous Data Sovereignty (IDS) and Governance (IDG):** These concepts assert the inherent right of Indigenous peoples to exercise ownership and governance over data about their communities, peoples, lands, resources, and knowledge systems, regardless of where the data is held or who collected it.[46] IDG refers to the right to autonomously decide how Indigenous data is collected, accessed, analyzed, interpreted, managed, and used, ensuring it reflects Indigenous priorities, values, and worldviews.[55] This challenges the tendency to extract Indigenous knowledge and analyze it solely within Western frameworks.[59]
- **OCAP® Principles:** Developed by First Nations in Canada, OCAP® (Ownership, Control, Access, Possession) provides a standard for research conduct.[57]
  - *Ownership:* Communities own their information collectively.[58]
  - *Control:* Communities control data collection, use, and management.[58]
  - *Access:* Communities must be able to access data about themselves and be involved in decisions about access.[58]
  - *Possession:* Physical control of data should ideally remain with the community or a designated steward.[58]
- **CARE Principles for Indigenous Data Governance:** Developed by the Global Indigenous Data Alliance (GIDA), these principles are people- and purpose-oriented, designed to complement the data-centric FAIR principles (Findable, Accessible, Interoperable, Reusable).[60]

- *Collective Benefit:* Data ecosystems must enable Indigenous peoples to derive benefit.[55]
- *Authority to Control:* Recognize Indigenous rights and empower their authority to control data.[55]
- *Responsibility:* Researchers have a responsibility to share how data is used to support self-determination and collective benefit, and to build capacity.[55]
- *Ethics:* Prioritize Indigenous rights and well-being, minimize harm, maximize benefit from Indigenous perspectives, ensure justice, and consider future use.[55]

Table 2: Principles for Indigenous Data Governance: CARE and FAIR

Principle Set	Core Tenets	Primary Focus	Key Goal	Relationship
FAIR	Findable, Accessible, Interoperable, Reusable	Data characteristics	Data Sharing / Reuse	Technical layer for data management
CARE	Collective Benefit, Authority to Control, Responsibility, Ethics	People / Purpose / Rights	Indigenous Self-determination / Benefit	Adds ethical/rights layer to FAIR's technical layer
Sources	56	60	55	56

- **Practical Guidelines:** Respectful engagement involves building meaningful relationships on the community's terms.[59] This includes acknowledging traditional territories [32], proper introductions (self-location) [32], respecting protocols around Elders and knowledge sharing [32], recognizing the limitations of translation across worldviews [59], keeping knowledge in context [59], using culturally appropriate sharing methods [59], returning data/materials upon request [51], ensuring Indigenous peoples control research within their territories [51], and ensuring collective benefit.[55]

Ethical engagement is not a procedural hurdle but a fundamental shift in research paradigm. It moves away from extractive models towards collaborative partnerships built on respect, reciprocity, and the recognition of Indigenous sovereignty over cultural heritage and knowledge. This requires researchers to relinquish unilateral control and become accountable partners, ensuring that any exploration of "latent algorithms" directly benefits and empowers the communities whose knowledge is being engaged.

Generative Threads: Contemporary Dialogues Between Weaving, Computation, and AI

The historical and conceptual links between weaving and computation continue to resonate in contemporary explorations involving generative art and artificial intelligence (AI). This emerging field presents both opportunities for cultural revitalization and significant ethical challenges, demanding careful navigation grounded in Indigenous protocols and perspectives.

Historical Links: Weaving and Computing

The connection between textiles and computation has deep roots. Ada Lovelace famously observed in 1843 that Charles Babbage's Analytical Engine "weaves algebraic patterns, just as the Jacquard loom weaves flowers and

leaves".[16] The Jacquard loom's use of punch cards to control intricate patterns operated on a binary-like system, a direct precursor to early computing technology.[16] This historical interweaving provides context for current explorations using AI in textile design and analysis.

AI/Generative Art Inspired by Indigenous Patterns

Contemporary artists and researchers are exploring the intersection of Indigenous weaving traditions and computational methods:

- **Generative Systems:** Projects utilize generative algorithms and AI to create new designs inspired by traditional rulesets and aesthetics.[16] This can involve training neural networks on datasets of traditional patterns (e.g., Indian and Islamic tile patterns, as a related example) or developing generative processes based on weaving logic (e.g., birch band weaving inspiring architectural installations [64]).
- **Cultural Interfaces:** Some projects explicitly bridge cultural heritage and digital technology. Guillermo Bert's "Encoded Textiles" project involved Mapuche weavers incorporating QR codes, visually similar to Indigenous graphic designs, into their textiles. Scanning the codes links viewers to Mapuche cultural narratives, transforming a potentially opaque symbol into a transmitter of cultural values.[17]
- **AI-Assisted Design:** Researchers propose AI-enabled pipelines for textile design that integrate historical pattern studies, mathematical modeling, and craft knowledge (like lacemaking).[16] AI could potentially optimize patterns for aesthetic appeal, cultural relevance, mechanical properties (like elasticity), or even integration with electronics.[16] AI might also assist in scaling complex traditional designs, such as zero-waste patterns like kimonos, which are difficult to adapt manually.[65]

"Ancestral Algorithms" and AI

A critical discourse is emerging around AI, reframing it through non-Western and ancestral knowledge systems:

- **Challenging AI Epistemologies:** Concepts like "Ancestral Immediacies" or "Ancestral Algorithms" position ancestrality itself as a technology of collective knowledge, challenging the linear, progress-oriented, and often colonial logics embedded in mainstream AI development.[45] These frameworks consult circular or quantum time concepts and emphasize embodied, relational ways of knowing, questioning algorithmic certainty.[66]
- **Decolonizing AI:** Research explores how Indigenous knowledge frameworks (e.g., Gullah Geechee) can inform AI development, redressing epistemological biases in data and algorithms and advancing anticolonial scholarship.[45] This involves questioning the singular cultural definition of "intelligence" underpinning much AI.[45]
- **AI, Ancestry, and Memory:** AI tools like Deep Nostalgia animate old photos, allowing users to "see" ancestors, raising complex questions about algorithmic representation, affect, memory, and the mediation of ancestral communication.[68] Other AI applications like image colorization also engage with historical representation but are susceptible to biases within their training data and algorithms.[69]

Bridging Heritage and Generative Aesthetics: Potential and Pitfalls

The use of AI and generative art holds potential but requires careful consideration:

- **Preservation and Evolution:** Generative AI could potentially support the preservation and evolution of textile heritage crafts by acting as a design resource, assisting ideation, simulating outcomes, or helping document and analyze patterns.[16]

- **Augmentation vs. Replacement:** The goal should be to augment the skills of artisans, not replace them.[16] AI tools should be developed *with* craftspeople, incorporating their embodied knowledge and respecting the value of human skill and learning communities.[16] AI is currently limited in its understanding of material properties (like fabric drape under stress) and the nuanced, embodied knowledge of master weavers.[65]
- **Ethical Imperatives:** The ease with which AI can generate or manipulate patterns significantly increases the risk of appropriation, superficial engagement, and misrepresentation of Indigenous cultural heritage. Using Indigenous patterns or "ancestral algorithms" in AI or generative art without deep, respectful, consent-based collaboration guided by Indigenous protocols (FPIC, IDS, CARE) risks perpetuating colonial dynamics of extraction and control.[16]

The intersection of AI, generative art, and Indigenous weaving represents a critical juncture. While offering potential pathways for cultural continuity and novel expression, it simultaneously carries profound risks if approached without a strong ethical foundation grounded in Indigenous sovereignty, relationality, and co-production of knowledge. The development and application of these powerful technologies must be guided not just by technical feasibility but by the principles of respect, reciprocity, and responsibility towards the knowledge holders and their heritage.

Conclusion: Re-Centering Relationships – Weaving as Co-Produced Knowledge

This exploration into the "latent algorithms" of Indigenous weaving networks reveals a profound intersection of cultural practice, mathematical thinking, and information management that long predates digital computation. The analysis, guided by lenses of computational patterning, oral knowledge, decolonial critique, circulation, and AI translation, highlights several key findings. Firstly, Pacific and Andean textiles demonstrably embody sophisticated mathematical concepts, including complex symmetries, iterative and recursive pattern generation, and intricate systems of information encoding like the khipu.[6] Secondly, the transmission of this complex knowledge relied on holistic systems integrating oral tradition, embodied practice, and mnemonic devices, deeply embedded in social and ecological relationships.[7] Thirdly, archaeological and ethnographic evidence indicates that this knowledge was not static or isolated but circulated dynamically across precolonial networks through trade, kinship, and migration.[1]

However, the investigation also underscores the critical limitations and potential dangers of applying purely formal, Western analytical frameworks to these traditions. A decolonial perspective reveals that concepts like "algorithm" or attempts at formalization risk reducing rich, contextualized practices to abstract logic, potentially overlooking or devaluing the relational, spiritual, and embodied dimensions central to Indigenous epistemologies.[11] The act of formalizing, if not undertaken with extreme care and collaboration, can become an extension of colonial epistemic violence.[12] This directly addresses the central tension of this inquiry: listening *for logic* must be secondary to, and guided by, listening *to relationships*. Attempting to formalize a cultural practice into code risks losing its soul – the web of connections, meanings, and lived experiences it embodies – if the relational context is ignored.

Therefore, ethical engagement is paramount. Protocols demanding Free, Prior, and Informed Consent (FPIC) and frameworks like Indigenous Data Sovereignty (IDS), OCAP, and the CARE principles (Collective Benefit, Authority to Control, Responsibility, Ethics) are not optional additions but essential foundations for any research involving Indigenous knowledge.[51] This necessitates a shift towards viewing knowledge as co-produced through respectful partnership, rather than extracted or "discovered" by external researchers.[46]

The emerging field of generative art and AI engagement with Indigenous patterns and "ancestral algorithms"

presents both exciting possibilities and heightened ethical stakes.[16] While AI might offer tools for cultural revitalization and new creative expression, it also magnifies the potential for appropriation and misrepresentation if not guided by Indigenous governance and collaborative principles.[16]

Limitations to this understanding persist, including the vast amount of knowledge potentially lost due to colonization and the inherent difficulties of translating across fundamentally different knowledge systems and worldviews.[29] Future research should prioritize deeply collaborative projects co-designed with Indigenous artists, mathematicians, and cultural technologists. Developing culturally-grounded computational tools that respect Indigenous protocols and epistemologies, and further exploring Indigenous philosophies of technology and pattern, represent vital directions.

Ultimately, understanding the "latent algorithms" of Indigenous weaving is not about finding a hidden, abstract code waiting to be deciphered by Western computational methods. It is about appreciating the intricate, intelligent, relational, and profoundly human (and more-than-human) systems of knowledge that Indigenous peoples have woven into the very fabric of their cultures for millennia. The goal is not to extract and formalize, but to engage in respectful dialogue, foster reciprocal learning, and honor the enduring wisdom woven into these textile traditions.

Works cited

1. Andean Textiles - The Metropolitan Museum of Art, accessed on April 19, 2025, https://www.metmuseum.org/essays/andean-textiles
2. WEAVING THE HARMONY OF CULTURES INTO THE INFINITE TAPESTRY OF MATHEMATICS | ARACÊ - newsciencepubl, accessed on April 19, 2025, https://periodicos.newsciencepubl.com/arace/article/view/584
3. Ethnomathematics: A Multicultural View of Mathematical Ideas 0412989417, 9780412989414 - DOKUMEN.PUB, accessed on April 19, 2025, https://dokumen.pub/ethnomathematics-a-multicultural-view-of-mathematical-ideas-0412989417-9780412989414.html
4. Redalyc.(Ethno)mathematics as discourse, accessed on April 19, 2025, https://www.redalyc.org/pdf/2912/291238322003.pdf
5. (PDF) (Ethno)mathematics as discourse - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/316001118_Ethnomathematics_as_discourse
6. The Question of Symmetry in Andean Textiles, accessed on April 19, 2025, https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1323&context=tsaconf
7. Design research with the use of visual and symmetry analysis in ..., accessed on April 19, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC9901921/
8. Inside-out frieze symmetries in ancient Peruvian weavings, accessed on April 19, 2025, https://www.math.stonybrook.edu/~tony/whatsnew/oct08/peru.html
9. Incan oral traditions | World Literature I Class Notes - Fiveable, accessed on April 19, 2025, https://fiveable.me/world-literature-i/unit-10/incan-oral-traditions/study-guide/mwcdv79fREmDlEIa
10. Weaving Wisdom in the Andes: Documentary explores the relationship between

an ancient craft and brain health - Dancers' Group, accessed on April 19, 2025, https://dancersgroup.org/2022/06/weaving-wisdom-in-the-andes/
11. Failures Indigenising school mathematics: A narrative inquiry - ERIC, accessed on April 19, 2025, https://files.eric.ed.gov/fulltext/EJ1458987.pdf
12. DHQ: Digital Humanities Quarterly: Voluptuous Disintegration: A Future History of Black Computational Thought, accessed on April 19, 2025, https://digitalhumanities.org/dhq/vol/16/3/000634/000634.html
13. "What's Happening": Applying Indigenous Epistemologies to Ecology - Scalar, accessed on April 19, 2025, https://scalar.oberlincollegelibrary.org/decolonizing-ethnomusicology/whats-happening-applying-indigenous-epistemologies-to-ecology
14. Lapita cultural complex - Science Learning Hub, accessed on April 19, 2025, https://www.sciencelearn.org.nz/resources/3205-lapita-cultural-complex
15. Lapita culture - Wikipedia, accessed on April 19, 2025, https://en.wikipedia.org/wiki/Lapita_culture
16. Generative AI for Textile Engineering: Blending Tradition and Functionality Through Lace, accessed on April 19, 2025, https://mit-genai.pubpub.org/pub/nz0prx2u
17. Weaving a Storytelling Interface : 8 Steps (with Pictures) - Instructables, accessed on April 19, 2025, https://www.instructables.com/Weaving-a-Storytelling-Interface/
18. Pacific Ethnomathematics: A Bibliographic Study - UH Press, accessed on April 19, 2025, https://uhpress.hawaii.edu/title/pacific-ethnomathematics-a-bibliographic-study/
19. Symmetry in Northwest Coast Spruce Root Basketry patterns, accessed on April 19, 2025, https://coe.hawaii.edu/ethnomath/wp-content/uploads/sites/12/2023/02/JamesJasmine_Lesson2.pdf
20. Ethnomathematics: An analysis of frieze and crystallographic patterns on Ulos, accessed on April 19, 2025, https://aip.scitation.org/doi/abs/10.1063/5.0113269
21. The mathematical ideas involved in Maranao Weaving - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/318964462_The_mathematical_ideas_involved_in_Maranao_Weaving
22. Symmetrical Design Analysis of Malay Arts in Ethnomathematics Context - Human Resource Management Academic Research Society, accessed on April 19, 2025, https://hrmars.com/papers_submitted/7984/symmetrical-design-analysis-of-malay-arts-in-ethnomathematics-context.pdf
23. Hiapo (tapa) - Smarthistory, accessed on April 19, 2025, https://smarthistory.org/hiapo-tapa/
24. Hiapo (tapa) (article) | Polynesia - Khan Academy, accessed on April 19, 2025, https://www.khanacademy.org/humanities/ap-art-history/pacific-apah/polynesia-apah/a/hiapo
25. South American Indigenous Textiles - Hatching Dragons, accessed on April 19,

2025, https://www.hatching-dragons.com/en-gb/blog/south-american-indigenous-textiles
26. (PDF) From Ethnomathematics to Ethnocomputing - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/279362099_From_Ethnomathematics_to_Ethnocomputing
27. (PDF) Indigenous Andean Voices at the Interface between the Oral ..., accessed on April 19, 2025, https://www.researchgate.net/publication/386278053_Indigenous_Andean_Voices_at_the_Interface_between_the_Oral_and_the_Written_The_Peruvian_Quechua_Traditions_from_Colonial_Huarochiri_ca_1608
28. Quipu - Wikipedia, accessed on April 19, 2025, https://en.wikipedia.org/wiki/Quipu
29. Full article: A Numerical Connection Between Two Khipus - Taylor & Francis Online, accessed on April 19, 2025, https://www.tandfonline.com/doi/full/10.1080/00776297.2024.2411789
30. Quipu - Inka knots, accessed on April 19, 2025, https://stateofjeffersonrotary.org/library/programs-1/1329-quipu-inka-knots.html
31. The Khipu (knot what you think...) - YouTube, accessed on April 19, 2025, https://www.youtube.com/watch?v=xaTjF5HAct8
32. Indigenous Protocols - Indigenous Research - University of Alberta Library Subject Guides, accessed on April 19, 2025, https://guides.library.ualberta.ca/indigenous-research/indigenous-data-sovereignty/indigenous-protocols
33. INDIGENOUS LANGUAGE RESEARCH, THE INSIDER–OUTSIDER EXPERIENCE AND WEAVING ABORIGINAL - UVic Journal Publishing Service, accessed on April 19, 2025, https://journals.uvic.ca/index.php/winhec/article/view/20293/9080
34. (PDF) Whatuora: Theorizing "New" Indigenous Research Methodology from "Old" Indigenous Weaving Practice - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/331393469_Whatuora_Theorizing_New_Indigenous_Research_Methodology_from_Old_Indigenous_Weaving_Practice
35. Shaped by the Loom: Weaving Worlds in the American Southwest - Bard Graduate Center, accessed on April 19, 2025, https://www.bgc.bard.edu/shaped-by-the-loom
36. Weaving in the Andes Mountains - THE KINDCRAFT, accessed on April 19, 2025, https://thekindcraft.com/weaving-in-the-andes-mountains/
37. The arts of memory : Comparative perspectives on a mental artifact | HAU, accessed on April 19, 2025, https://www.journals.uchicago.edu/doi/10.14318/hau2.2.025
38. Why Knot? - DSpace@MIT, accessed on April 19, 2025, https://dspace.mit.edu/bitstream/handle/1721.1/27981/24577962-MIT.pdf?sequence=2&isAllowed=y
39. Placing diverse knowledge systems at the core of transformative climate research - PMC, accessed on April 19, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10406791/

40. Talepakemalai: Lapita and Its Transformations in the Mussau Islands of Near Oceania, accessed on April 19, 2025, https://ioa.ucla.edu/talepakemalai
41. Deconstructing the Lapita Cultural Complex in the Bismarck Archipelago - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/261567052_Deconstructing_the_Lapita_Cultural_Complex_in_the_Bismarck_Archipelago
42. Oceanic Tattooing and the Implied Lapita Ceramic Connection - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/303873240_Oceanic_tattooing_and_the_implied_Lapita_ceramic_connection
43. Tapa Cloth - Museum of Natural and Cultural History, accessed on April 19, 2025, https://mnch.uoregon.edu/collections-galleries/tapa-cloth
44. Representing Reason: Feminist Theory and Formal Logic, accessed on April 19, 2025, http://fitelson.org/piksi/representing_reason.pdf
45. Ancestors and Algorithms: Ethnocomputing AI with African & African Diasporic Heritage, accessed on April 19, 2025, https://deepblue.lib.umich.edu/handle/2027.42/174511
46. spectrum.library.concordia.ca, accessed on April 19, 2025, https://spectrum.library.concordia.ca/986506/7/Indigenous_Protocol_and_AI_2020.pdf
47. Indigenous Knowledge and Knowledge Synthesis, Translation and Exchange (KSTE), accessed on April 19, 2025, https://www.nccih.ca/docs/context/RPT-IndigenousKnowledgeKSTE-Ellison-EN.pdf
48. Decolonizing Mathematics Instruction: Subordinating Teaching to Learning - SciELO, accessed on April 19, 2025, https://www.scielo.br/j/bolema/a/pVLFvgGPbx3TPhZ3MSc4TTn/?lang=en
49. An Interrogation of Sensory Anthropology of and in Japan - Monash University, accessed on April 19, 2025, https://researchmgt.monash.edu/ws/files/284492523/270808520.pdf
50. Social Experience and Anthropological Knowledge - CiteSeerX, accessed on April 19, 2025, https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=a4720a7e64500c580725de3219ad5f53a76c63c5
51. Principles & Guidelines for the Protection of the Heritage of Indigenous People | Alaska Native Knowledge Network, accessed on April 19, 2025, https://www.uaf.edu/ankn/indigenous-knowledge-syst/principles-guidelines-for/
52. English, accessed on April 19, 2025, https://ich.unesco.org/doc/src/U022-v2.0-HO1-EN__Free_prior_and_informed_consent.docx
53. African Unit 6: Respecting Indigenous Knowledge Systems - Michigan State University, accessed on April 19, 2025, https://aac.matrix.msu.edu/modules/african-modules/african-unit-6-respecting-indigenous-knowledge-systems/
54. Ethics and Intangible Cultural Heritage, accessed on April 19, 2025,

https://ich.unesco.org/en/ethics-and-ich-00866
55. CARE Principles for Indigenous Data Governance, accessed on April 19, 2025, https://www.rd-alliance.org/wp-content/uploads/2024/03/CARE20Principles20for20Indigenous20Data20Governance_OnePagers_FINAL20Sept2006202019.pdf
56. Indigenous Knowledge and Indigenous Data Sovereignty - Research support - The University of Queensland, accessed on April 19, 2025, https://research-support.uq.edu.au/indigenous-knowledge-and-indigenous-data-sovereignty
57. Indigenous Data Sovereignty - Indigenous Research - Subject Guides at University of Alberta Libraries, accessed on April 19, 2025, https://guides.library.ualberta.ca/indigenous-research/indigenous-data-sovereignty
58. Indigenous Data Sovereignty - Research at UCalgary - University of Calgary, accessed on April 19, 2025, https://research.ucalgary.ca/engage-research/indigenous-research-support-team/irst-resources/indigenous-data-sovereignty
59. Beyond Conservation: Working Respectfully with Indigenous People and Their Knowledge Systems, accessed on April 19, 2025, https://ipcaknowledgebasket.ca/resources/working-respectfully-with-indigenous-people-and-their-knowledge-systems/
60. CARE Principles for Indigenous Data Governance | ARDC, accessed on April 19, 2025, https://ardc.edu.au/resource/the-care-principles/
61. CARE Principles for Indigenous Data Governance - Wikipedia, accessed on April 19, 2025, https://en.wikipedia.org/wiki/CARE_Principles_for_Indigenous_Data_Governance
62. CARE Principles — Global Indigenous Data Alliance, accessed on April 19, 2025, https://www.gida-global.org/care
63. Generative Art and Design for Re-rendering the Handicrafts industry, accessed on April 19, 2025, https://generativeart.com/reserved_to_participants/15_Jill_Shah.pdf
64. GA2017 – XX Generative Art Conference Page 1 TITLE The Light Keeper, Ceiling Installation (Paper and Poster) Topic: Architect, accessed on April 19, 2025, https://www.generativeart.com/GA2017/Manuel%20Baez%20GA2017_abstract%20and%20paper.pdf
65. Artificial intelligence in textile patterns and sizing - Specialty Fabrics Review, accessed on April 19, 2025, https://specialtyfabricsreview.com/2024/09/01/artificial-intelligence-in-textile-patterns-and-sizing/
66. AI: Ancestral Immediacies | HKW Haus der Kulturen der Welt, accessed on April 19, 2025, https://www.hkw.de/en/programme/ancestral-immediacies
67. Ancestral Protocols and Technologies of Conviviality | HKW Haus der Kulturen der Welt, accessed on April 19, 2025, https://www.hkw.de/en/programme/ancestral-immediacies/ancestral-protocols-and-technologies-of-conviviality
68. Full article: The Talking (Virtual) Dead: Ancestral Communication And/As

Technology, accessed on April 19, 2025, https://www.tandfonline.com/doi/full/10.1080/15267431.2025.2485911?src=
69. On AI colourisation: algorithms, ancestry, and colour beyond the black box - ResearchGate, accessed on April 19, 2025, https://www.researchgate.net/publication/389067698_On_AI_colourisation_algorithms_ancestry_and_colour_beyond_the_black_box
70. Indigenous knowledge systems and language practice - Sabinet African Journals, accessed on April 19, 2025, https://journals.co.za/doi/pdf/10.10520/AJA16844998_49